

Session 1: Multiple linear regression review

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CUNY SPH Biostatistics 2

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Learning objectives and outline

Multiple
Linear
Regression

Interaction
(effect
modification)

Analysis of
Variance

Model
formulae

Learning objectives and outline

Learning objectives

- 1 identify systematic and random components of a multiple linear regression model
- 2 define terminology used in a multiple linear regression model
- 3 define and explain the use of dummy variables
- 4 interpret multiple linear regression coefficients for continuous and categorical variables
- 5 use model formulae to multiple linear models
- 6 define and interpret interactions between variables
- 7 interpret ANOVA tables

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- 1 multiple regression terminology and notation
- 2 continuous & categorical predictors
- 3 interactions
- 4 ANOVA tables
- 5 Model formulae

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Multiple Linear Regression

Systematic part of model

For more detail: Vittinghoff section 4.2

$$E[y|x] = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_p x_p$$

- $E[y|x]$ is the expected value of y given x
- y is the outcome, response, or dependent variable
- x is the vector of predictors / independent variables
- x_p are the individual predictors or independent variables
- β_p are the regression coefficients

Random part of model

$$y_i = E[y_i|x_i] + \epsilon_i$$

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \dots + \beta_p x_{pi} + \epsilon_i$$

- x_{ji} is the value of predictor x_j for observation i

Assumption: $\epsilon_i \stackrel{iid}{\sim} N(0, \sigma_\epsilon^2)$

- Normal distribution
- Mean zero at every value of predictors
- Constant variance at every value of predictors
- Values that are statistically independent

Continuous predictors

- **Coding:** as-is, or may be scaled to unit variance (which results in *adjusted* regression coefficients)
- **Interpretation for linear regression:** An increase of one unit of the predictor results in this much difference in the continuous outcome variable
 - *additive model*

Binary predictors (2 levels)

- **Coding:** indicator or dummy variable (0-1 coding)
- **Interpretation for linear regression:** the increase or decrease in average outcome levels in the group coded “1”, compared to the reference category (“0”)
 - e.g. $E(y|x) = \beta_0 + \beta_1 x$
 - where $x = \{ 1 \text{ if male, } 0 \text{ if female } \}$

Multilevel Categorical Predictors (Ordinal or Nominal)

- **Coding:** $K - 1$ dummy variables for K -level categorical variables *
- **Interpretation for linear regression:** as above, the comparisons are done with respect to the reference category
- Testing significance of multilevel categorical predictor: partial F-test, a.k.a. nested ANOVA

* STATA and R code dummy variables automatically, behind-the-scenes

Inference from multiple linear regression

- Coefficients are t-distributed when assumptions are correct
- Variance in the estimates of each coefficient can be calculated
- The t-test of the null hypothesis $H_0 : \beta_1 = 0$ and from confidence intervals tests whether x_1 predicts y , *holding other predictors constant*
 - often used in causal inference to control for confounding: see section 4.4

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Interaction (effect modification)

How is interaction / effect modification modeled?

Interaction is modeled as the product of two covariates:

$$E[y|x] = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 * x_2$$

What is interaction / effect modification?

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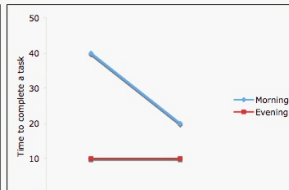
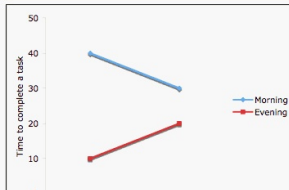
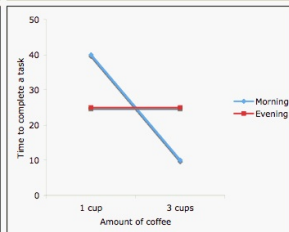
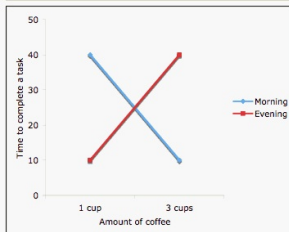
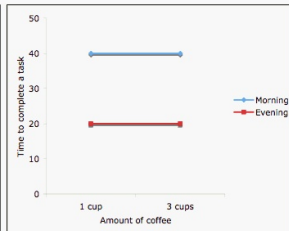
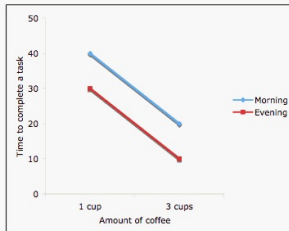
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Analysis of Variance

Review of the ANOVA table

Source of Variation	Sum Sq	Deg Fr	Mean Sq	F
Model	MSS	k	MSS/k	(MSS/k)/
Residual	RSS	n-(k-1)	RSS/(n-k-1)	
Total	TSS	n-1		

- k = Model degrees of freedom = coefficients - 1
- n = Number of observations
- **F** is F-distributed with k numerator and $n - (k - 1)$ denominator degrees of freedom

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Model formulae

What are model formulae?

Model formulae tutorial

- Model formulae are shortcuts to defining linear models in R
- Regression functions in R such as `aov()`, `lm()`, `glm()`, and `coxph()` all accept the “model formula” interface.
- The formula determines the model that will be built (and tested) by the R procedure. The basic format is:
response variable ~ explanatory variables
- The tilde means “is modeled by” or “is modeled as a function of.”

Model formula for simple linear regression

$$y \sim x$$

- where “x” is the explanatory (independent) variable
- “y” is the response (dependent) variable.

Model formula for multiple linear regression

Additional explanatory variables would be added as follows:

$$y \sim x + z$$

Note that “+” does not have its usual meaning, which would be achieved by:

$$y \sim I(x + z)$$

Types of standard linear models

`lm( y ~ u + v )`

u and v factors: **ANOVA**

u and v numeric: **multiple regression**

one factor, one numeric: **ANCOVA**

Model formulae cheatsheet

symbol	example	meaning
+	+ x	include this variable
-	- x	delete this variable
:	x : z	include the interaction
*	x * z	include these variables and their interaction
/	x / z	nesting: include z nested within x
	x z	conditioning: include x given z
^	(u + v + w)^3	include these variables and all interactions up to three way
1	-1	intercept: delete the intercept

Model formulae comprehension Q&A #1

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How to interpret the following model formulae?

$$y \sim u + v + w + u:v + u:w + v:w$$

$$y \sim u * v * w - u:v:w$$

$$y \sim (u + v + w)^2$$

Model formulae comprehension Q&A #2

How to interpret the following model formulae?

$$y \sim u + v + w + u:v + u:w + v:w + u:v:w$$

$$y \sim u * v * w$$

$$y \sim (u + v + w)^3$$